Lab 3 Submission Report

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Answer Part 1: Git Use with Engineering Workflow

- **GitHub Repository URL:** https://github.com/apfirst13/toktickit
- **Main Branch Strategy:** `main` (Production) ← `lab3-staging` (Integration) ← `feat/*` (Feature Branches).
- **Key Pull Requests:**
  - PR #59: Admin User Management (Issue 8)
  - PR #60: E2E Testing Full Flow (Issue 9)
  - PR #61: Visual Inspection & Screenshots (Issue 10)
  - PR #62: Release Integration & Documentation (Issue 11)

Answer Part 2: Spec DD

Documentation updated for Lab 3 specifications:

- `docs/lab-03/specification.md`: Business Rules and Functional Requirements.
- `docs/lab-03/api-spec.md`: API Endpoints (Auth, Staff, Admin).
- `docs/lab-03/ui-spec.md`: UI Wireframes and Constraints.

Answer Part 3: Test DD and Traceability

Traceability matrix and test specifications are documented in:

- `docs/lab-03/tests.md`: Covers ACs, BRs, Security, and Edge Cases.

Answer Part 4: AI Use with Reflection

AI usage prompts and formal reflection are documented in:

- `docs/lab-03/ai-use.md`: Contains 9+ actual prompts and analysis of AI strengths/limitations.

Answer Part 5: Auth & Role Management

Implemented secure JWT authentication, initial password changes, and database roles.

- **Route Guard:** `App.tsx` forces `mustChangePassword` check.
- **Tests:** `e2e/lab-03/authentication.spec.ts`

Answer Part 6: IT Staff Ticket Queue & Detail

IT Staff can view, filter, and add internal notes to tickets.

- **Internal Notes:** Golden tint applied, visible only to Staff/Admin.
- **Tests:** `e2e/lab-03/staff-ticket-flow.spec.ts`

Answer Part 7: Administrator User Management

Administrators can manage users with strict safety constraints (cannot deactivate self/last admin).

- **Safety Guards:** Handled in `users.controller.ts`.
- **Tests:** `e2e/lab-03/user-administration.spec.ts`

Answer Part 8: Peer Review Log

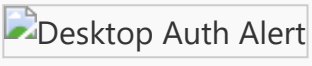
Cross-project peer review feedback and resolution are documented in:

- `docs/lab-03/reviewer.md`: Features feedback from `Theerawat82` and `sawaddeekubkuy-creator`.

Answer Part 9: Zen Green UI and Responsive Evidence (VIS-01)

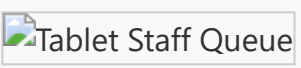
Automated visual inspection passing all breakpoints (Desktop, Tablet, Mobile) generated via Playwright.

Auth Alert



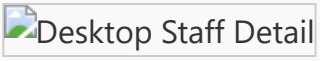
Desktop - Security Alert for Password Change

Staff Queue (Tablet)



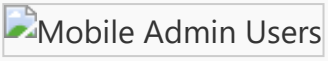
Tablet - IT Staff Ticket Queue (Responsive Card Layout)

Staff Detail (Desktop)



Desktop - IT Staff Ticket Detail with Internal Note

Admin Users (Mobile)



Mobile - Administrator User Management